

141 Measure space with your hands

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MINS

Astronomers measure the size of stars and planets and the distance between them in degrees. You can use your hand to measure degrees—“measuring space” as you do so.

What you need

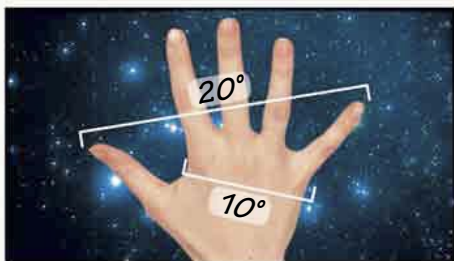
- Your hands
- A starry night



1 Start with your index fingertip. The width of this fingertip, held out at arm's length, is about 1 degree across.



2 Now curve this finger and hold it side-on, still at arm's length. Starting at the top, your three finger joints are equal to roughly 3, 4, and 6 degrees wide.



3 Finally, use your entire hand. The measure from your pinky fingertip to your thumb is about 20 degrees, while the width of your palm is about 10 degrees.

142 Make a moonscape

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Fill a baking sheet with sand. Smooth the sand with a ruler, then drop round objects into it. Try a grapefruit, a marble, a ball bearing, or a grape. Each represents a “meteor strike” on your moonscape.



Make different-sized craters

